

Data Science Project

Executive Summary

What the analysis does

Previously published research from Office for National Statistics (ONS) confirms that there is a high prevalence of adults who were previously Looked After Children who have since left care ('care leavers', abbreviated CL below) appearing before the Adult Criminal Courts.

Reports show that of the CL born in 1994, 52% had a criminal conviction on their record by the age of 24 compared with 13% of children who had not been in care (ONS, 2022).

15% had then received a custodial sentence - which is over 10 times the rate recorded for their peers. (ONS, 2022).

A further 37% of CL had received a non-custodial conviction, such as a financial penalty or Community Order, compared with 12% of children who had not been in care (ONS, 2022).

This project is not about recreating that finding, it instead looks at a factor that may contribute to it - what happens to an CL's housing situation once they leave the care system and does that influence their future? Does the data support the hypothesis that unsuitable accommodation or homelessness is a contributing factor to a criminal conviction?

Key findings

Published statistics from the Department for Education (2025) and the Ministry of Housing, Communities and Local Government (2025) were used, covering England from 2018 to 2025.

This data only captures the details of those CL that stayed in contact with the relevant services and is not a true reflection of every care-experienced adult.

Published reports show that the number of CL in Unsuitable Accommodation rose by 11% between 2022 and 2024 (Department for Education, 2025), whilst the number declared officially Homeless rose by 65% for the same time period (Ministry of Housing, Communities and Local Government, 2025).

The link to conviction rates cannot be achieved as there are no publicly available datasets showing convictions in relation to care status.

Outcome

The data does not support a direct link between housing risk and conviction, since no publicly available dataset connects the two.

It does show that homelessness among CL is rising faster than accommodation problems can explain, indicating housing instability is a growing risk in its own right.

Background

Obtaining publicly available records linking conviction status to care status was not possible at the start of the project. The underlying data behind the ONS research was never released publicly, so that study cannot be reproduced for analysis.

This changed the original hypotheses, which had set out to question whether CL status directly influenced conviction rates. The project was redefined around the datasets that are available and could be merged for analysis and so the scope was narrowed to the measurable subset of housing outcomes for CL.

Data Infrastructure and Tools

Data Sets:

The project uses two accredited sources, both published under the Open Government Licence:

1. Department for Education's (DfE) care leaver accommodation suitability tables
2. Ministry of Housing, Communities and Local Government's (MHCLG) statutory homelessness statistics

Both datasets were published at national, regional and Local Authority (LA) level. The downloads also have both LA and reporting year as shared identifiers

As the data references local authorities, comparison is possible so a location can be compared against the average for England alongside the national trends. Manchester was chosen because it has consistent data coverage across the study period.

Each source groups ages differently which means direct comparison is complicated.

- Source #1 reports ages 17 to 18 and 19 to 21 in bandings.
- Source #2 reports 18 to 20, 21 to 24 and 25+ in bandings.

As these are not directly comparable, and to ensure that there is no forced match, the two measures are treated as overlapping but separate throughout. This is noted wherever the data is compared directly

Suppressed values (marked 'c' where counts are too small to publish in Source #1) were identified and excluded from calculations. This is noted as a limitation of the underlying data, not a choice made in this project.

Reporting periods also differ slightly: DfE uses financial year to 31 March, while MHCLG reports both financial-year totals and quarterly breakdowns, requiring the quarterly data to be excluded in favour of matching annual figures.

There is no publicly available data that links care status to Adult Criminal Court conviction records at either age or geographical level. This is an issue with the UK's public data infrastructure in general, required data is not always available and is not specific to this project only. However, it is a limitation that shaped what the data analytics of this project can and cannot analyse

Tools:

- Excel for structure checks, quality validation and visualisation
- Python for automation and analysis
- Pandas for joining

Data Engineering

The datasets were downloaded in their published formats. They were then filtered for location before any transformation work started.

The data was standardised across both files using a common structure of columns: LA, year, and metric — this ensured that the two sources could be combined into one working table

The biggest challenge was the age banding mismatch between the datasets. Rather than forcing the two age bands together, they were kept as separate metrics. For the homelessness comparison specifically, CL were expressed as a percentage of all households owed a homelessness duty, which avoids needing the two age bands to align at all.

The full ETL script and its output are provided as Figures 1 to 4 below, so every transformation step can be checked against the original source

Quality checks were completed as follows:

- Manchester was confirmed as present in every reporting year of the accommodation dataset, 2021-2025. Its homelessness figures are only available from 2024 onward, as set out below.
- Suppressed values, where the original data was too small to publish, were identified and excluded from calculations.
- Manchester's homelessness figures before 2024 were blank in the source file, not suppressed, simply not published for that period, and were retained as genuine gaps rather than removed or estimated.
- Each dataset's published totals were reconciled against the filtered rows to confirm the transformation had not corrupted the original data

This resulted in a single table, indexed by LA and year, which was used for both the national trend and the Manchester comparison. Figures 1 to 4 below evidence this process, from the raw source data through to the final joined table.

[illegible]

FIGURE 1: EXAMPLE OF RAW SOURCE DATA FROM THE DfE DATASET, SHOWING A SUPPRESSED VALUE MARKED C FOR MANCHESTER, 2025.

	A	B	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	
1	Table A3 – Number of households owed a homelessness duty by support needs of household England, April 2024 to March 2025																		
2																			
3																			
4			Households with no support needs owed duty ^{1,2}	Households with unknown support needs owed duty	Number of households Households with one or more support needs owed duty ^{1,2}				Total number of support needs ¹	Of which households with:			Young parent requiring support to manage independently	Care leaver aged 18-20 years	Care leaver aged 21-24 ³	Care leaver aged 25+ ⁴	Care leaver - retired option ⁵		
5					Total households with support needs	1	2	3+		Young person aged 16-17 years	Young person aged 18-25 years requiring support to manage independently								
6	E92000001	ENGLAND	141,620	270	188,520	83,280	45,350	59,880	431,190	2,710	12,680	3,670	4,610	1,750	2,220		700		
7	E12000007	London	38,230	90	32,430	17,510	7,700	7,220	62,650	440	1,950	650	350	230	180		130		
165	E07000074	Maldon																	
166	E07000235	Malvern Hills	126	0	132	76	32	24	228	2	5	3	5	1	1		0		
167	E08000003	Manchester	3,085	0	1,616	768	382	466	3,532	3	177	18	36	17	24		8		
168	E07000174	Manfield	276	0	253	179	36	38	405	0	16	8	3	4	0		0		

FIGURE 2: EXAMPLE OF RAW SOURCE DATA FROM THE MHCLG DATASET, TABLE A3, SHOWING HOMELESSNESS SUPPORT NEEDS BY LOCAL AUTHORITY.

```

69 home = pd.concat([nat_home, manc_home], ignore_index=True)
70 home['homeless_pct_of_all'] = (home['homeless_count'] / home['total_owed'] * 100).round(2)
71
72
73 master = acc.merge(home, on=['year', 'area'], how='outer').sort_values(['area', 'year'])
74 output_path = Path('master_joined_table.csv').resolve()
75 master.to_csv(output_path, index=False)
76
77 print(f"Saved output successfully to: {output_path}")
78
79 print(master.to_string(index=False))
80

```

```

2021 Manchester      45          9      NaN      NaN      NaN
2022 Manchester      34          6      NaN      NaN      NaN
2023 Manchester      33          5      NaN      NaN      NaN
2024 Manchester      36          5      44.0    5879.0    0.75
2025 Manchester      43          6      36.0    4701.0    0.77

```

```

GLOBAL: /Apprenticeship/Professional Practice/Summative/Project 1 - Data
$ python "9-4413-bc7f-67279702a768/etl_pipeline.py"
• Saved output successfully to: C:\Apprenticeship\Professional Practice\Summative\Project 1 - Data\master_joined_table.csv

```

```

year area accom_unsuitable_count accom_unsuitable_pct homeless_count total_owed homeless_pct_of_all
2019 England NaN NaN 2790.0 269510.0 1.04
2020 England NaN NaN 3390.0 289800.0 1.17
2021 England 1740 5 3290.0 270560.0 1.22
2022 England 1850 6 3390.0 279470.0 1.21
2023 England 1950 6 3740.0 300910.0 1.24
2024 England 2050 6 4330.0 327410.0 1.32
2025 England 2040 5 4610.0 330410.0 1.40
2021 Manchester 45 9 NaN NaN NaN
2022 Manchester 34 6 NaN NaN NaN
2023 Manchester 33 5 NaN NaN NaN
2024 Manchester 36 5 44.0 5879.0 0.75
2025 Manchester 43 6 36.0 4701.0 0.77

```

FIGURE 3: THE ETL SCRIPT (PYTHON, USING PANDAS) AND ITS TERMINAL OUTPUT, SHOWING THE SCRIPT EXECUTING SUCCESSFULLY AND PRODUCING THE JOINED TABLE.

	A	B	C	D	E	F	G
1	year	area	accom_unsuitable_count	accom_unsuitable_pct	homeless_count	total_owed	homeless_pct_of_all
2	2019	England			2790	269510	1.04
3	2020	England			3390	289800	1.17
4	2021	England	1740	5	3290	270560	1.22
5	2022	England	1850	6	3390	279470	1.21
6	2023	England	1950	6	3740	300910	1.24
7	2024	England	2050	6	4330	327410	1.32
8	2025	England	2040	5	4610	330410	1.40
9	2021	Manchester	45	9			
0	2022	Manchester	34	6			
1	2023	Manchester	33	5			
2	2024	Manchester	36	5	44	5879	0.75
3	2025	Manchester	43	6	36	4701	0.77
4							

FIGURE 4: THE FINAL JOINED TABLE, SAVED AS MASTER JOINED TABLE DOT CSV.

Data Visualisation & Dashboards

The first chart shows the percentage of CL aged 19-21 recorded in unsuitable accommodation, comparing Manchester against England from 2021 to 2025.

A line chart was used because the visualisation is showing change over time, not a fixed comparison and a line makes the shape of that change visible in a way a bar chart would not.

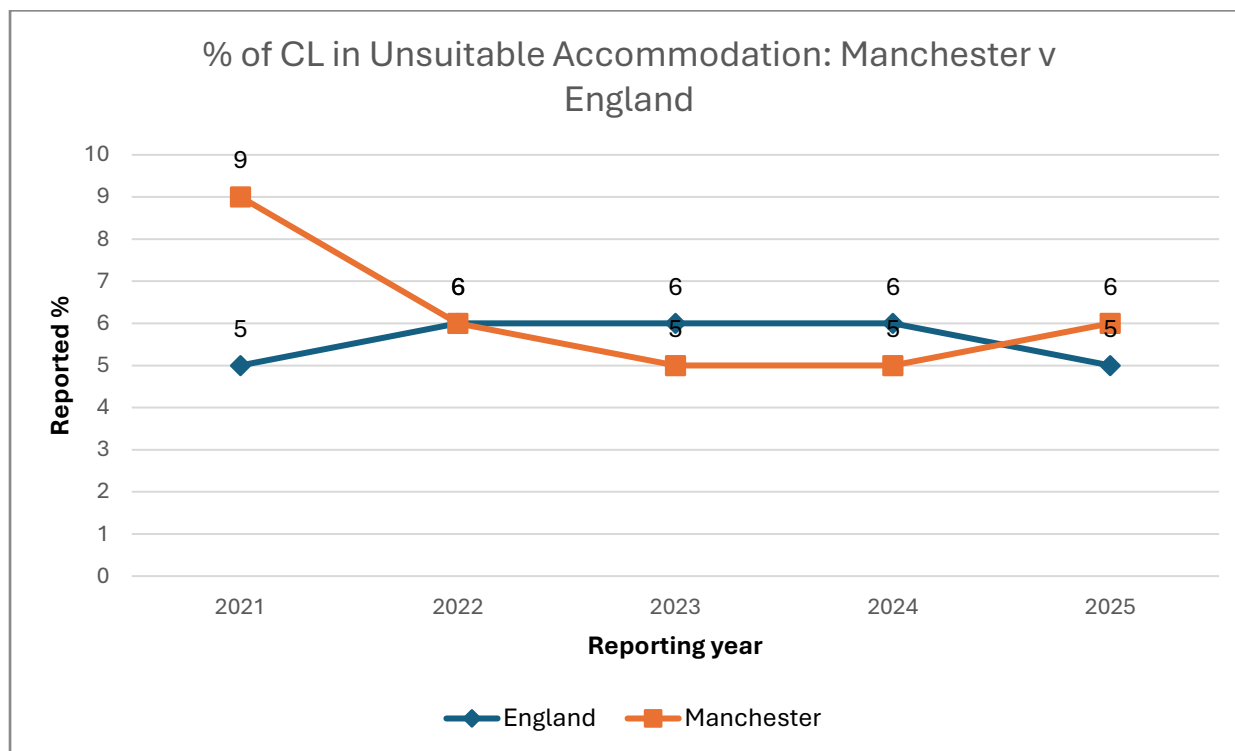


FIGURE 5: CL IN UNSUITABLE ACCOMMODATION, AGED 19-21, MANCHESTER VERSUS ENGLAND, 2021-2025.

The second chart compares CL as a share of all households owed a homelessness duty, Manchester against England, 2023-24 and 2024-25.

This was built as a slope chart rather than a clustered bar chart, since the two lines connecting a single start and end point make the direction of change immediately visible, which is the actual finding being made here.

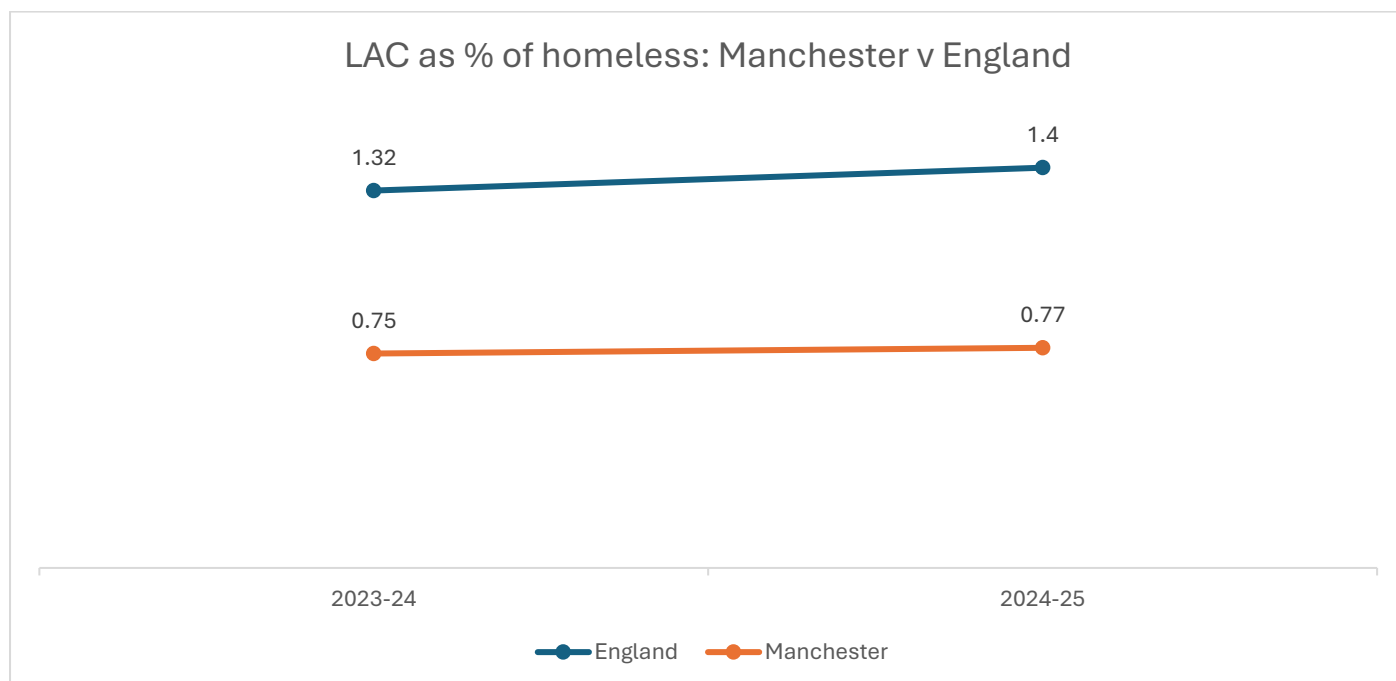


FIGURE 6: CL AS A SHARE OF ALL HOUSEHOLDS OWED A HOMELESSNESS DUTY, MANCHESTER VERSUS ENGLAND, 2023-24 AND 2024-25.

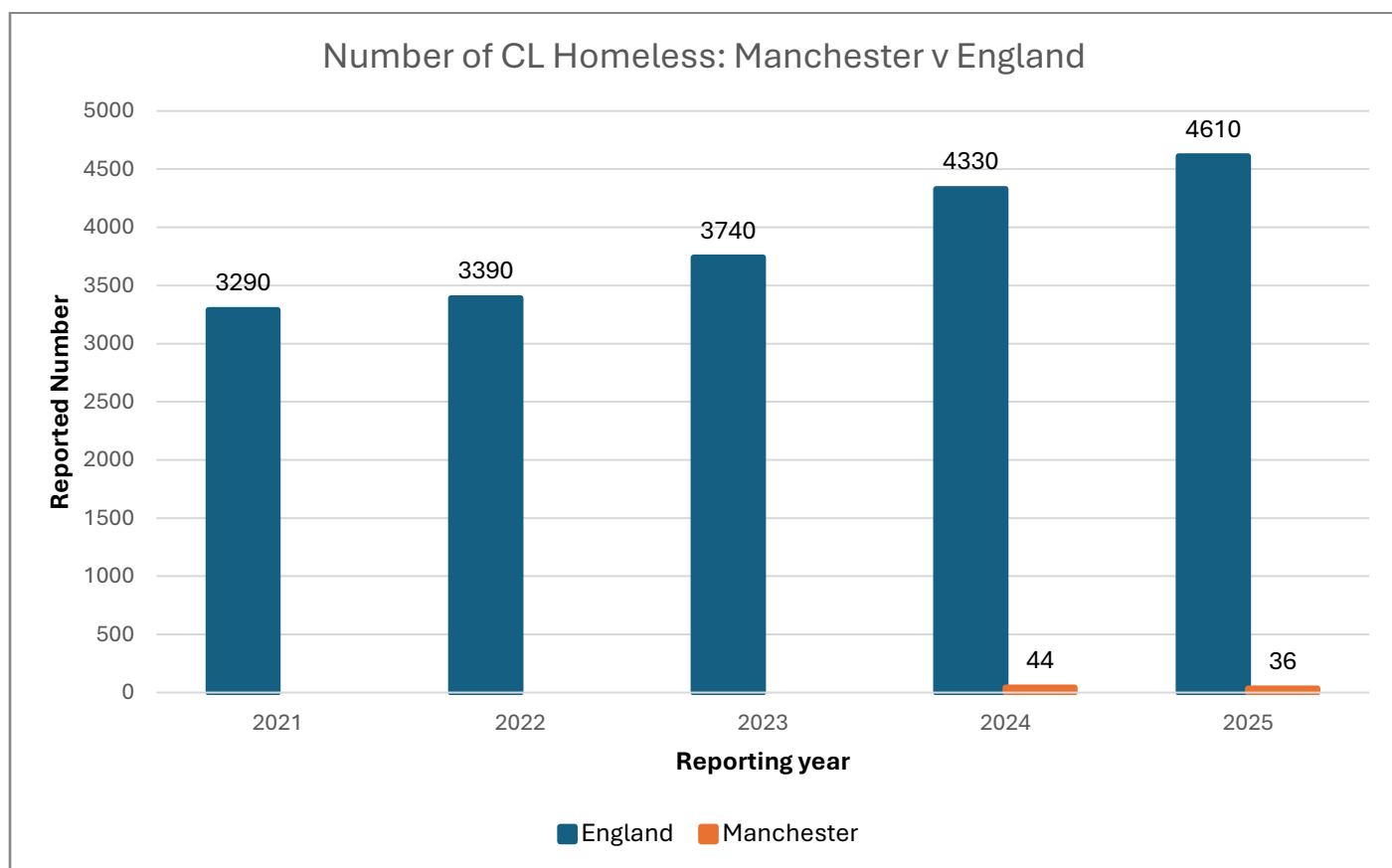


FIGURE 7: CL IN UNSUITABLE ACCOMMODATION, RAW COUNTS, MANCHESTER VERSUS ENGLAND, 2021-2025.

The third chart shows the raw count of CL in unsuitable accommodation, Manchester against England, on a single axis with no percentage conversion.

This was included specifically to show scale: Manchester's numbers are genuinely small next to England's national total, and showing this as raw counts rather than percentages makes that scale difference honest and immediate, rather than hidden behind two similar-looking percentage figures.

Manchester's share of England's total unsuitable-accommodation count is proportionate to its share of the national CL population so the size difference reflects population scale rather than Manchester performing disproportionately worse or better than the national picture.

A corresponding chart showing raw homelessness counts is provided in the appendix for reference, alongside the percentage figures used in the main analysis above.

Data Analytics

Hypothesis

The rate of CL in unsuitable accommodation and the rate of CL recorded as homeless rose together between 2018 and 2025, both nationally and in Manchester specifically.

H1 is rejected if the two measures show clearly different rates of change, since that would mean the two datasets are picking up on different things rather than one shared pattern.

Method

A descriptive comparison of the two time series was used, at both national and Manchester level. There are not enough annual data points for a formal statistical test, and no natural control group exists. Percentage change over the available years was calculated for each measure. No causal claim follows from this.

Findings

H1 is rejected. Nationally, unsuitable accommodation among 19-21 year old CL rose by around 11% between 2022 and 2024. Homelessness among 18-20 year old CL rose by around 65% between 2018-19 and 2024-25, close to three times the rate.

In Manchester, the accommodation figure tracked England closely across all five years available, both areas sitting between 5% and 9% throughout. Manchester's homelessness count actually fell between 2023-24 and 2024-25 (44 to 36), while England's rose over the same period. This looked at first like a local improvement, but Manchester's share of its own total homelessness caseload stayed almost flat (0.75% to 0.77%), because Manchester's overall homelessness numbers dropped by roughly 20% in that year. The fall in care leaver numbers reflects a citywide drop, not something specific to CL.

Ethics

Only aggregated, published statistics are used. No individual-level data is processed. Both datasets only capture CL who remain in contact with statutory services, so those who lose contact do not appear in either measure. This gap could be checked against the Department for Education's separate "not in touch" figures, which were not used in this analysis. Including them in future work would show whether the true scale of the problem is larger than what this project has found.

Recommendations

Future work should prioritise obtaining a longer run of Manchester's homelessness data, since only two reporting years were available for this project against five years for accommodation.

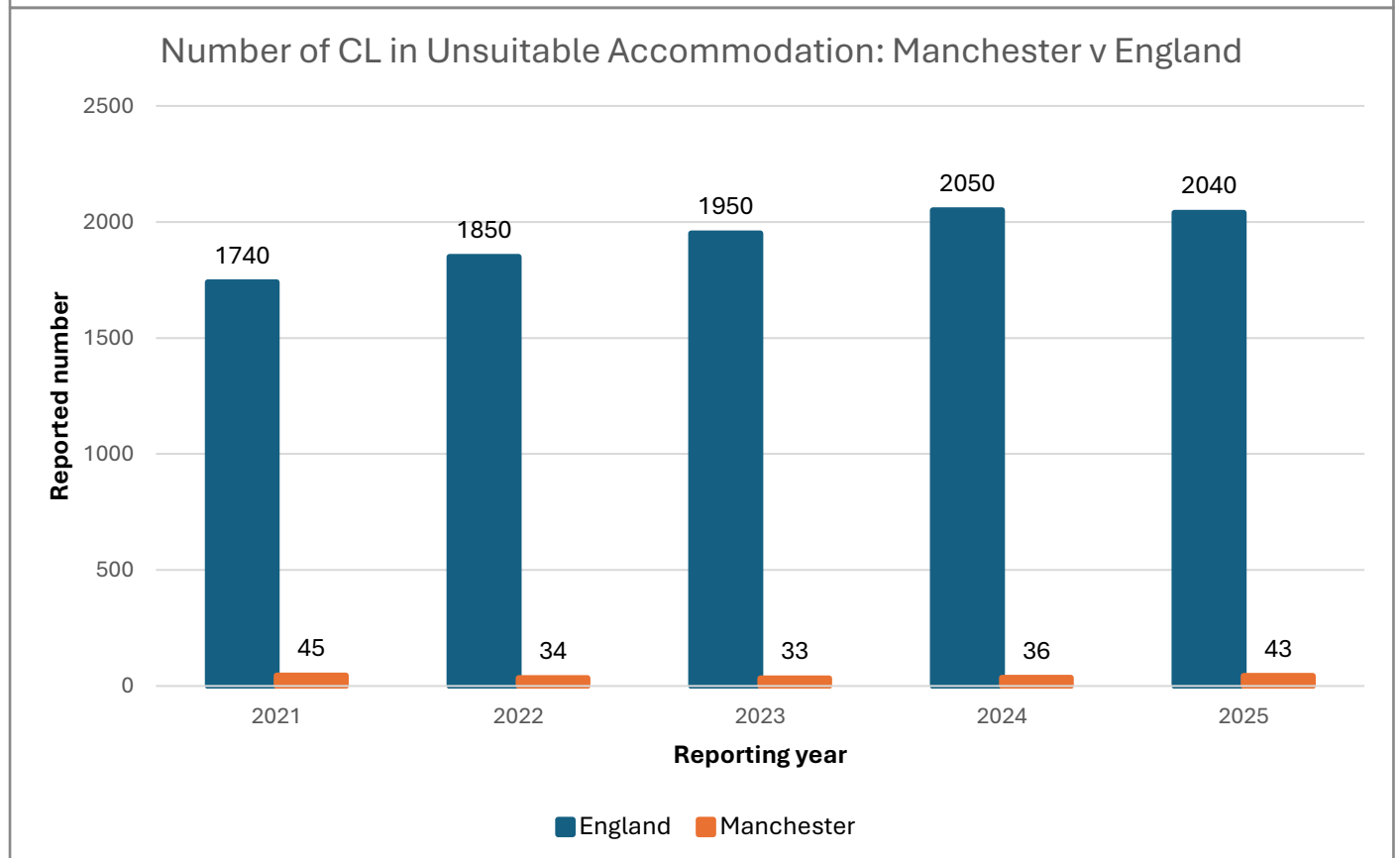
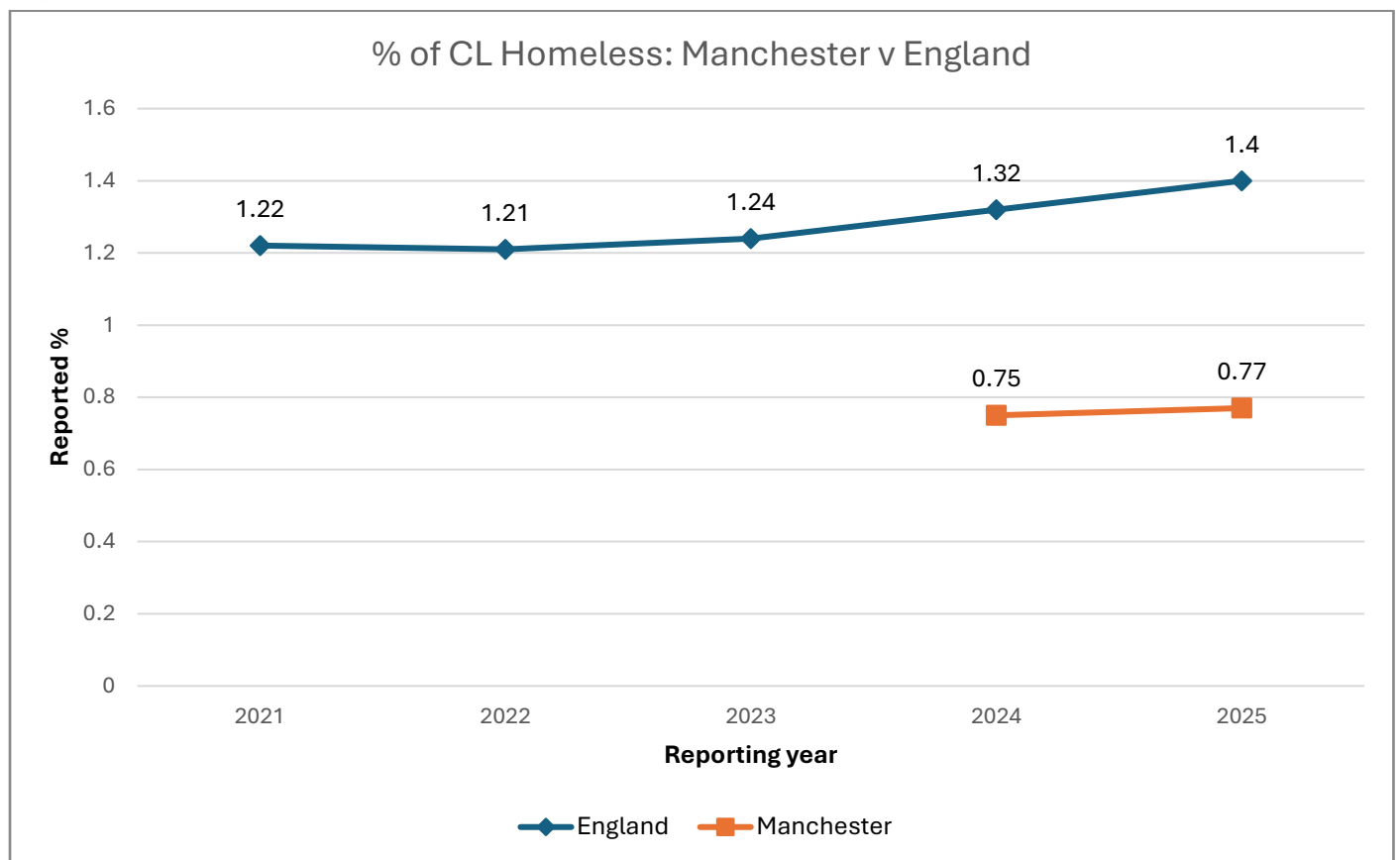
A single additional annual file would allow a genuine local trend to be compared against the national one, rather than a two-point snapshot.

Testing this analysis against other local authorities beyond Manchester would show whether the patterns found here are specific to one city or hold more broadly.

Manchester was chosen for consistent data coverage, not because it was expected to be representative.

If a future government data-linkage project makes record-level data available, similar to the approach used internally by ONS for its 2022 study, this would allow the housing and conviction datasets to be joined directly, rather than treated as separate, un-combined evidence as they are here.

Appendix



References

Department for Education (2025) Children looked after in England including adoptions: care leavers, whether their accommodation is suitable, by local authority.

Available at: <https://explore-education-statistics.service.gov.uk/data-catalogue/data-set/0048e808-4a62-4bcf-9d09-2e9a3968127b>

(Accessed: 22/08/2026).

Ministry of Housing, Communities and Local Government (2025) Statutory homelessness live tables.

Available at: <https://www.gov.uk/government/statistical-data-sets/live-tables-on-homelessness>

(Accessed: 22/08/2026).

Office for National Statistics (2022) The education background of looked-after children who interact with the criminal justice system.

Available at:

<https://www.ons.gov.uk/peoplepopulationandcommunity/educationandchildcare/articles/theeducationbackgroundoflookedafterchildrenwhointeractwiththecriminaljusticesystem/december2022>

(Accessed: 22/08/2026).